

Parsa Eskandar

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SUMMARY

PhD candidate in computational genomics at the UC Santa Cruz Genomics Institute (Paten lab). I build scalable algorithms and high-performance C++ systems for pangenome graphs: compressed full-text indexes, haplotype-aware read mapping, and pangenome-based variant calling. First author at WABI 2025 and *Algorithms for Molecular Biology*; co-author of *Personalized Pangenome References (Nature Methods)*, *Pangenome-aware DeepVariant*, and HPRC2, in ongoing collaboration with Google's DeepVariant team. My index now powers a pangenome search and lift-over service in the UCSC Genome Browser.

EDUCATION

University of California, Santa Cruz

PhD Candidate, Biomolecular Engineering and Bioinformatics

Anticipated graduation: 06/28. Advisor: Prof. Benedict Paten.

GPA: 4.0/4.0

09/2022 – present

Sharif University of Technology

B.Sc. Computer Engineering

GPA: 3.47/4.0

09/2017 – 03/2022

RESEARCH & WORK EXPERIENCE

Graduate Student Researcher

UC Santa Cruz Genomics Institute (Computational Genomics Lab, Paten lab)

Santa Cruz, CA

09/2022 – present

- **Pangenome search and lift-over service for the UCSC Genome Browser (2026).** Designed and built a C++ web service over the full HPRC Release 2 graph (464 haplotypes, 2.6 Tbp). One query searches every haplotype at once, and lift-over between any two haplotypes is computed on demand from the graph rather than from precomputed pairwise chains. 23 GiB sampled index; a gene lifts over in ~0.1 s. Live on the development browser ahead of public release; presented as a talk and poster at HPRC 2026.
- **Lossless pangenome indexing.** Designed and implemented in C++ a run-length-compressed index that extends the r-index to report every graph position of a query with no loss of information, with a memory-efficient construction that builds chromosomes independently and merges them into a whole-genome index. Open source: github.com/parsaeskandar/pangenome-index. First-author papers at WABI 2025 and *Algorithms for Molecular Biology* 2026.
- **Personalized pangenome references (Nature Methods 2024).** Second author. Led the variant-calling evaluation of personalized pangenomes against standard linear-reference pipelines on GIAB samples and integrated them into the Giraffe-DeepVariant workflow, in collaboration with Google's DeepVariant team.
- **Pangenome-aware DeepVariant and HPRC Release 2.** Contributed to the UCSC-Google collaboration that adds pangenome haplotypes to DeepVariant (up to 25.5% fewer errors than linear-reference DeepVariant); contributing author on HPRC2.
- **Open-source vg toolkit.** Contributor to the vg pangenome toolkit: maintain its Giraffe-DeepVariant workflows and documentation. Taught a hands-on Giraffe + DeepVariant pangenome variant-calling workshop at ASHG 2025.

Computational Sciences and Informatics Intern

Roche Diagnostics

06/2025 – 09/2025

- Developed methods for allele-specific copy number calling from sequencing data; resulting method is the subject of pending patent application PCT/US2025/044005 (co-inventor).
- Built computational workflows for downstream analysis of large-scale sequencing datasets in a multidisciplinary team.

Research Assistant / Data Analyst

Sharif University of Technology & AI-Med Institute

Tehran, Iran

2020 – 2022

- B.Sc. thesis: high-frequency trading algorithms for the Iranian stock market (Prof. Fazli's lab).
- Real-time ECG segmentation with CNNs on records from >70,000 patients (Prof. Rabiee's lab, with Tehran University of Medical Sciences).
- Data analyst at AI-Med Institute, applying machine learning to clinical cohort data for digital-health services.

PUBLICATIONS

1. **P. Eskandar**, B. Paten, J. Sirén. "Lossless pangenome indexing using tag arrays." *Algorithms for Molecular Biology*, 21:12 (2026). doi:10.1186/s13015-026-00301-4.
2. J. K. Lucas, P. Hebbbar, W.-W. Liao, ..., **P. Eskandar**, ..., B. Paten (Human Pangenome Reference Consortium). "HPRC2: A human pangenome reference with near-complete coverage of common genetic variation." *bioRxiv* 2026.07.21.739710 (2026). doi:10.64898/2026.07.21.739710.
3. **P. Eskandar**, B. Paten, J. Sirén. "Lossless Pangenome Indexing Using Tag Arrays." **25th International Workshop on Algorithms in Bioinformatics (WABI 2025)**, LIPIcs vol. 344, pp. 8:1–8:20, Schloss Dagstuhl (2025). doi:10.4230/LIPIcs.WABI.2025.8.
4. M. Asri, P.-C. Chang, J. C. Mier, J. Sirén, **P. Eskandar**, ..., A. Carroll, B. Paten, K. Shafin. "Pangenome-aware DeepVariant." *bioRxiv* 2025.06.05.657102 (2025). doi:10.1101/2025.06.05.657102.
5. J. Sirén, **P. Eskandar**, M. T. Ungaro, G. Hickey, J. M. Eizenga, A. M. Novak, X. Chang, P.-C. Chang, M. Kolmogorov, A. Carroll, J. Monlong, B. Paten. "Personalized pangenome references." *Nature Methods* 21, 2017–2023 (2024). doi:10.1038/s41592-024-02407-2.

PATENTS

- Co-inventor, international patent application **PCT/US2025/044005** (pending), "Methods and Systems for Allele-Specific Copy Number Calling," assigned to Roche Diagnostics.

AWARDS AND HONORS

- Iran National Elite Foundation member, 2015 – 2022
- Bronze medal, Iran National Mathematical Olympiad, 2015
- Ranked 176th of ~150,000 in the national university entrance exam (mathematics), 2017

TECHNICAL SKILLS

Languages: C/C++ (primary), Python, Java, R, MATLAB, Bash

Algorithms: succinct and compressed data structures, string and graph algorithms, pangenome graphs

Genomics: read mapping and variant calling pipelines (vg Giraffe, DeepVariant, GATK, DRAGEN), GIAB benchmarking

Tools: Git, Linux, Slurm, Docker, WDL, Snakemake, PyTorch, TensorFlow, CMake, HTTP/JSON services